COSMIC CALCULATION SERIES / ORIGINAL PRACTICE

Celestial Sphere

02.01 · Original club practice problems

Show your reasoning, use SI units where appropriate, and state any assumptions. Constants are supplied in the questions that need them.

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These are original MAIS PAC practice exercises, not official IOAA problems.

PROBLEMS

01 Altitude at culmination

An observer at latitude 48° N observes a star of declination +20°. Assume a spherical Earth and ignore refraction. (a) Find the star’s altitude at upper culmination using h = 90° − |latitude − declination|. (b) Is the star north or south of the zenith at upper culmination? (c) Explain why its hour angle is zero at that instant.

02 Circumpolar skies

An observer is at 48° N. Ignore atmospheric refraction and local obstructions. (a) Find the minimum declination for a star to remain above the horizon throughout its daily motion, allowing the limiting case to touch the horizon. (b) Determine whether stars at declinations +50° and +30° are circumpolar. (c) Which declinations never rise above the horizon?

03 Right ascension and hour angle

The local sidereal time is 10 h 20 min. A star has right ascension 8 h 50 min. (a) Find its hour angle using H = LST − RA. (b) Is it east or west of the local meridian? (c) How long ago did it cross the meridian, in sidereal time? State your sign convention.

04 Seasons on the celestial sphere

The Earth’s rotation axis is tilted relative to the normal to its orbital plane. (a) Describe the celestial equator and the ecliptic. (b) Why does the Sun’s declination change during the year? (c) Explain the meaning of an equinox and a solstice. Do not assume the Earth–Sun distance causes the seasons.